

COURSE SLIDES · WSQ

Business Innovation with Internet-of-Things (IoT)

WSQ Course Code: TGS-2026064711

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v10.1 · 11 August 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

MORNING (AM)
Scan on arrival

2

AFTERNOON (PM)
Scan after lunch

3

ASSESSMENT
Scan before you begin

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How Digital Attendance Works

1

Mandatory for WSQ funding

AM, PM and Assessment attendance must all be taken for WSQ-funded courses.

2

The QR code

The trainer or administrator displays the QR code generated from the SSG portal.

3

How to submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

At least 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE
Internet of Things, digital transformation and business innovation.

DELIVERS
WSQ courses on IoT, AI, data analytics and emerging digital technology.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- How your organisation uses (or could use) connected devices and sensor data today.
- One business problem you would like IoT to solve by the end of today.

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HOUSEKEEPING

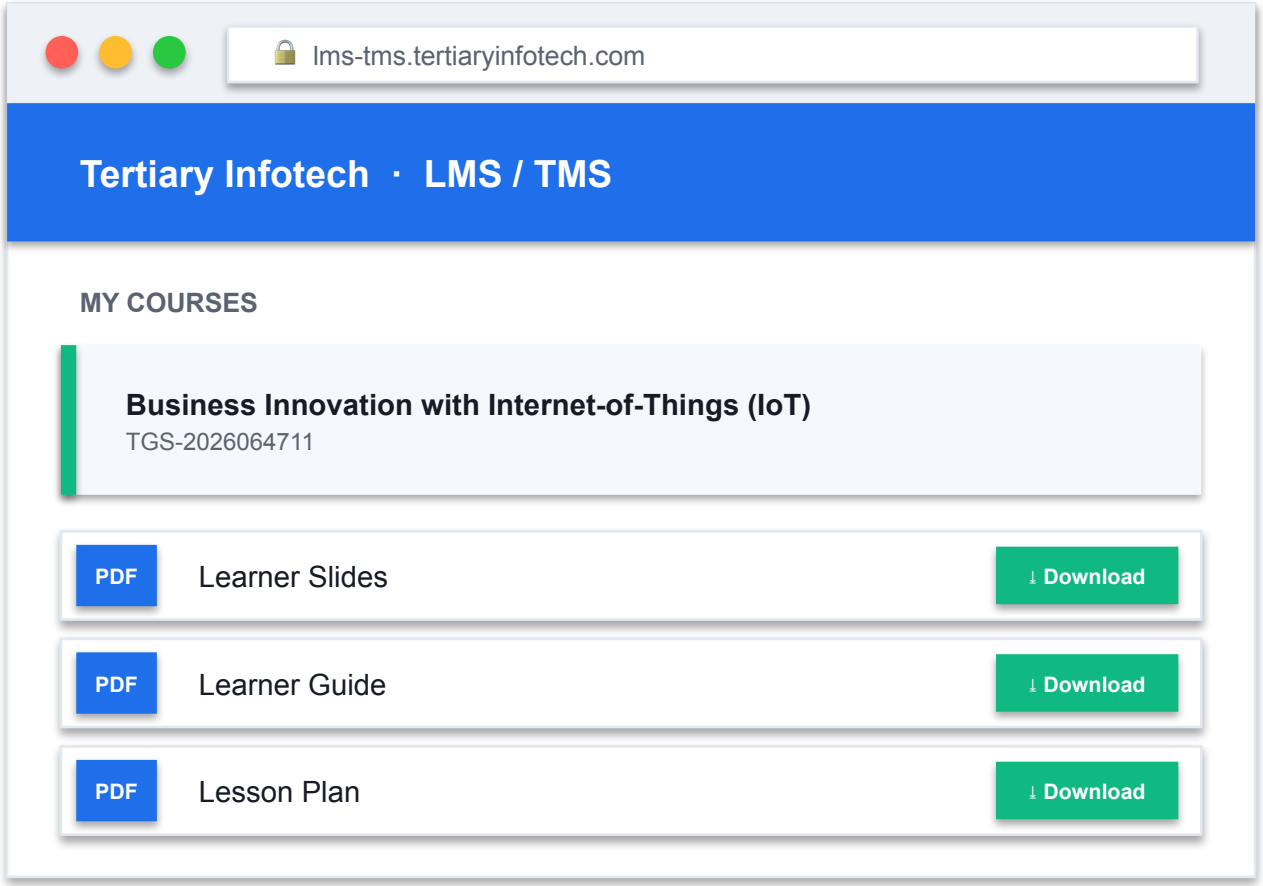
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL

Download Your Course Material



1. **Go to lms-tms.tertiaryinfotech.com**
   Open the portal in any browser.
2. **Log in with your registered email**
   The same address used to enrol.
3. **Open this course**
   Business Innovation with IoT.
4. **Download the material**
   Slides, Learner Guide and Lesson Plan.
5. **Keep them open in class**
   The assessment is open book.

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Skills Framework — TSC

1

TSC Title
Business Innovation

2

TSC Code
ICT-SNA-4003-1.1

3

Category
Business and Project Management

4

Proficiency
Level 4 — Explore opportunities for business innovation and reform, and lead the implementation of innovative business initiatives.

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TSC Abilities — Assessed by the Written Assessment

1. **A1**
   Compare current business model for the organisation with the other business models in the industry

2. **A2**
   Explore potential opportunities for business innovation to be introduced within the organisation

3. **A3**
   Identify ways in which digitisation can be applied to the business

4. **A4**
   Conduct feasibility analysis and weigh the costs-benefits of potential business innovation opportunities

5. **A5**
   Implement business innovation processes in line with a defined action plan

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TSC Knowledge — Assessed by Oral Questioning

1

K1

New and emerging innovative business processes / models deployed in the industry

2

K2

Concept and principles of digitisation

3

K3

Factors to consider to assess the viability of business innovation for a given organisation context

4

K4

Success factors for implementation of business innovation

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SCHEDULE · 9:30am – 6:30pm

Lesson Plan — 1 Day, 8 hours

Morning · 9:30am – 12:30pm

- Digital Attendance (AM) · Introductions
- Topic 1: Overview of IoT (LO1)
- Topic 2: Potential Opportunities of IoT (LO2)
- Topic 3: IoT Technology — concepts (LO3)
- Lunch Break — 1 hour

Afternoon · 1:30pm – 6:30pm

- Digital Attendance (PM)
- Topic 3: Hands-On Practical — Labs 1–4 (90 min)
- Topic 4: IoT Adoption (LO4)
- Topic 5: IoT Implementation (LO5)
- TRAQOM Survey · Digital Attendance (Assessment)
- Final Assessment — WA (75 min) + OQ (15 min)

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Compare business models

Compare your current business model with emerging IoT-enabled models. (A1, K1)

2

LO2 · Explore opportunities

Explore potential opportunities for business innovation using IoT. (A2, K1)

3

LO3 · Identify how IoT works

Identify how IoT works and how digitisation applies to the business. (A3, K2)

4

LO4 · Weigh costs & benefits

Conduct a feasibility analysis and weigh the costs and benefits of IoT. (A4, K3)

5

LO5 · Implement with a plan

Implement business innovation processes with IoT to a defined action plan. (A5, K4)

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FIVE TOPICS · ONE DAY

Course Outline

1

Topic 1 — Overview of IoT

Industry 4.0, IoT business models, sensors and actuators, wireless technologies.

2

Topic 2 — Potential Opportunities

Agriculture, autonomous vehicles, logistics, smart buildings, healthcare, drones.

3

Topic 3 — IoT Technology

Cloud platforms, posting data, remote control, analytics, automation + 4 hands-on labs.

4

Topic 4 — IoT Adoption

Adoption challenges, IoT security, edge computing, cost-benefit analysis.

5

Topic 5 — IoT Implementation

Success factors, design thinking, a 7-step implementation action plan.

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Briefing for Assessment

1. Place phones and other materials under the table or on the floor.
2. No photos or recording of assessment scripts.
3. No discussion during the assessment.
4. Use a black or blue pen for hard-copy assessments.
5. No liquid paper or correction tape.
6. Scripts are collected when the time is up.

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ASSESSMENT · 90 MINUTES TOTAL

Final Assessment

WA (Q&A) — 75 min

- Short-answer questions
- Assesses abilities A1–A5
- 15 minutes per ability
- Open book

OQ — 15 min

- Oral questioning
- Assesses knowledge K1–K4
- 1:1 with the assessor
- Open book

What you may use

- The course slides
- Your Learner Guide
- Approved materials only
- An appeal process is available

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit the WA (Q&A) — 75 min, open book
4. Oral Questioning — 15 min, 1:1 with the assessor
5. Sign the Assessment Summary Record

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01

TOPIC 01

Overview of Internet of Things (IoT)

Industry 4.0 · IoT business models · Sensors & actuators · Wireless technologies

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WHY NOW

We are in the 4th Industrial Revolution.

AI + IoT + 5G + Cloud + Robotics are converging — physical assets now report their own state in real time.

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LO1 · A1, K1 · 60 min

Key Concepts — Overview of Internet of Things (IoT)

1

Industry 4.0

IoT, AI, 5G, cloud and robotics converge so physical assets report their own state in real time.

2

What IoT is

Connecting everyday objects to the internet to cut cost, lift productivity and open new revenue.

3

The four building blocks

Sensors/devices capture data, connectivity moves it, cloud processes it, a UI acts on it.

4

Why now

Cheap sensors, long-range low-power wireless and 5G removed the cost barrier to mass adoption.

5

Product to service

IoT shifts firms from one-off product sales to subscription, outcome and pay-per-use models.

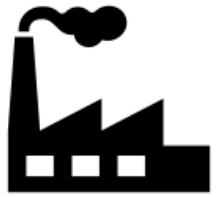
6

Data as the asset

The recurring value is the telemetry stream, not the device — it compounds over time.

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The Fourth Industrial Revolution



Industry 1.0
Mechanization, steam power, weaving loom

(18th century, 1760 - 1840)



Industry 2.0
Mass production, assembly line, electrical energy

(19th century, 1871 - 1914)



Industry 3.0
Automation, computers and electronics

(late 20th century, 1970 - present)



Industry 4.0
Cyber Physical Systems, internet of things, networks

(early 21st century, 2011 - present)

Industry 4.0 — cyber-physical systems, IoT, cloud computing and cognitive computing.

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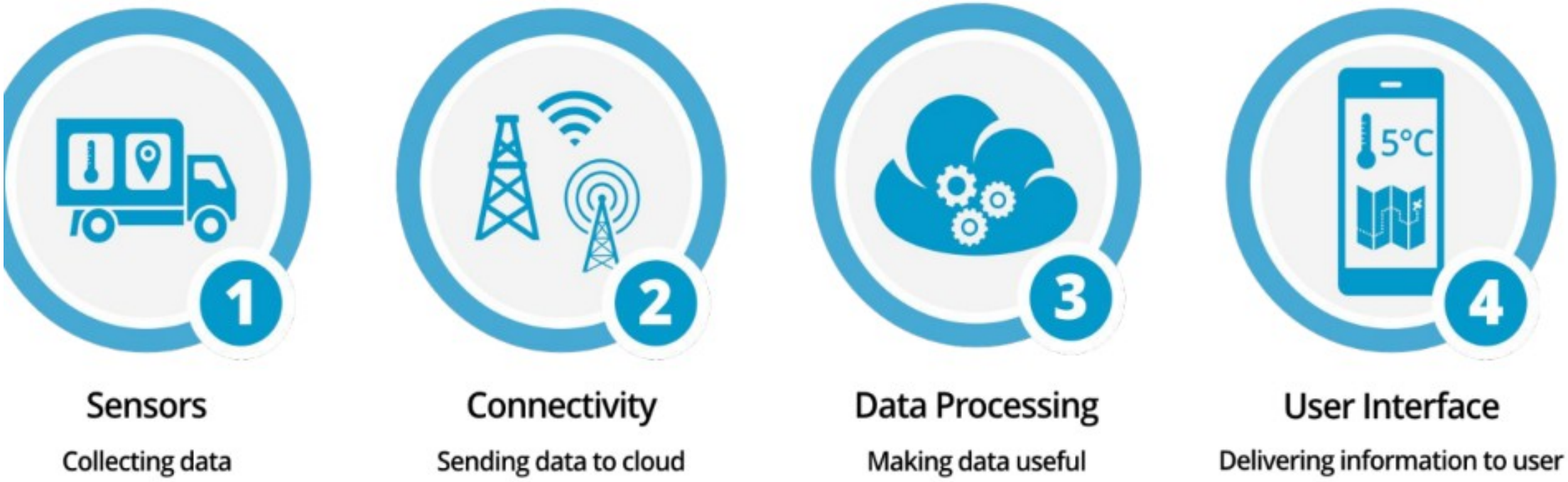
What Is the Internet of Things?



IoT connects everyday objects to the internet so they can send and receive data.

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The Components of an IoT System



A 'thing' is any object with embedded electronics that transfers data over a network.

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THE OPPORTUNITY

The Scale of IoT

$9T

Estimated IoT market value

50B+

Connected devices worldwide

4th

Industrial revolution underway

5G

The accelerator to mass adoption

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How IoT Works — The Four Building Blocks

1

SENSORS & DEVICES capture data from the physical world

2

CONNECTIVITY carries it over WiFi, cellular, LoRa or MQTT

3

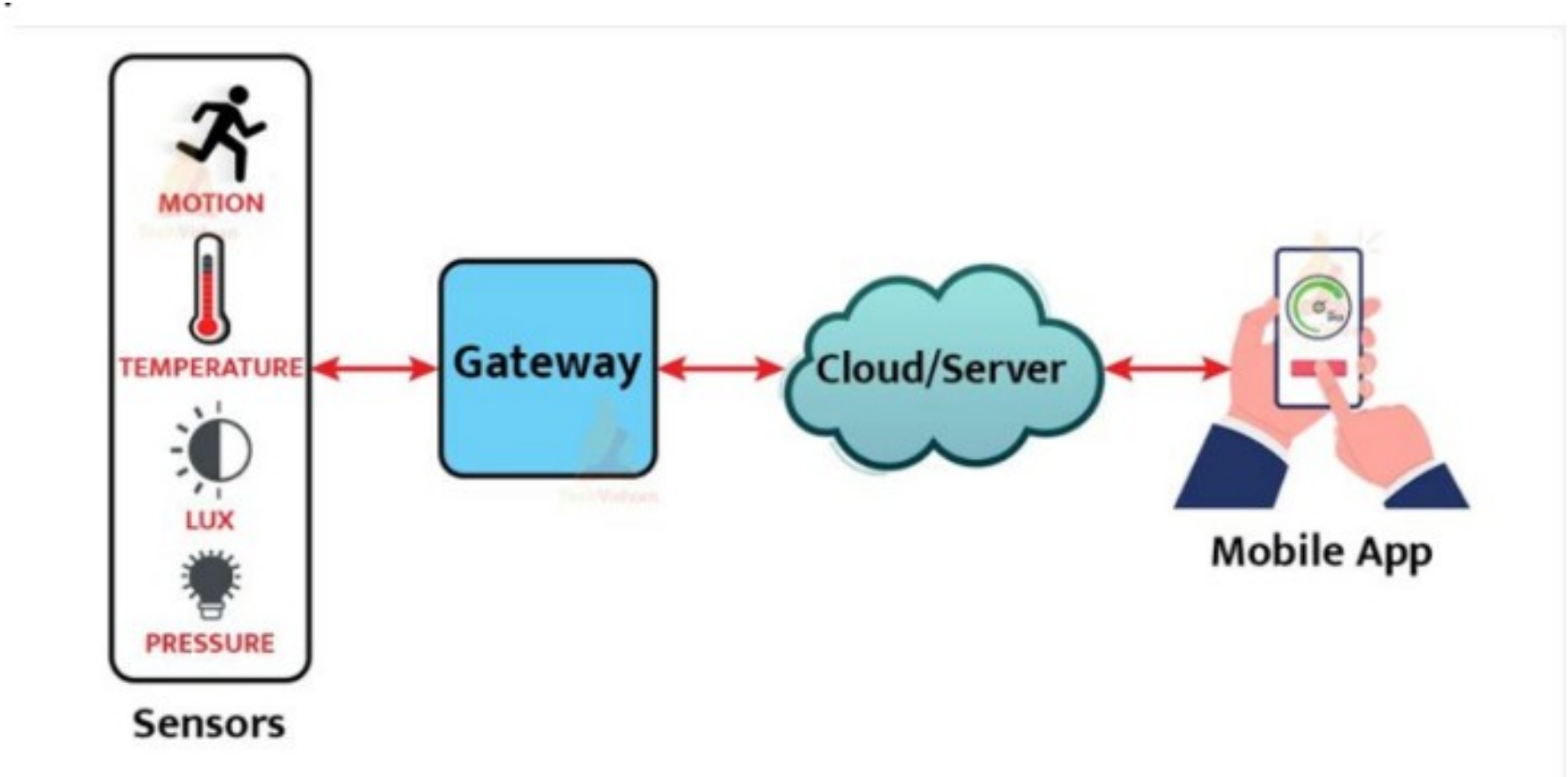
CLOUD stores, processes and analyses the data

4

USER INTERFACE shows insight and sends commands back

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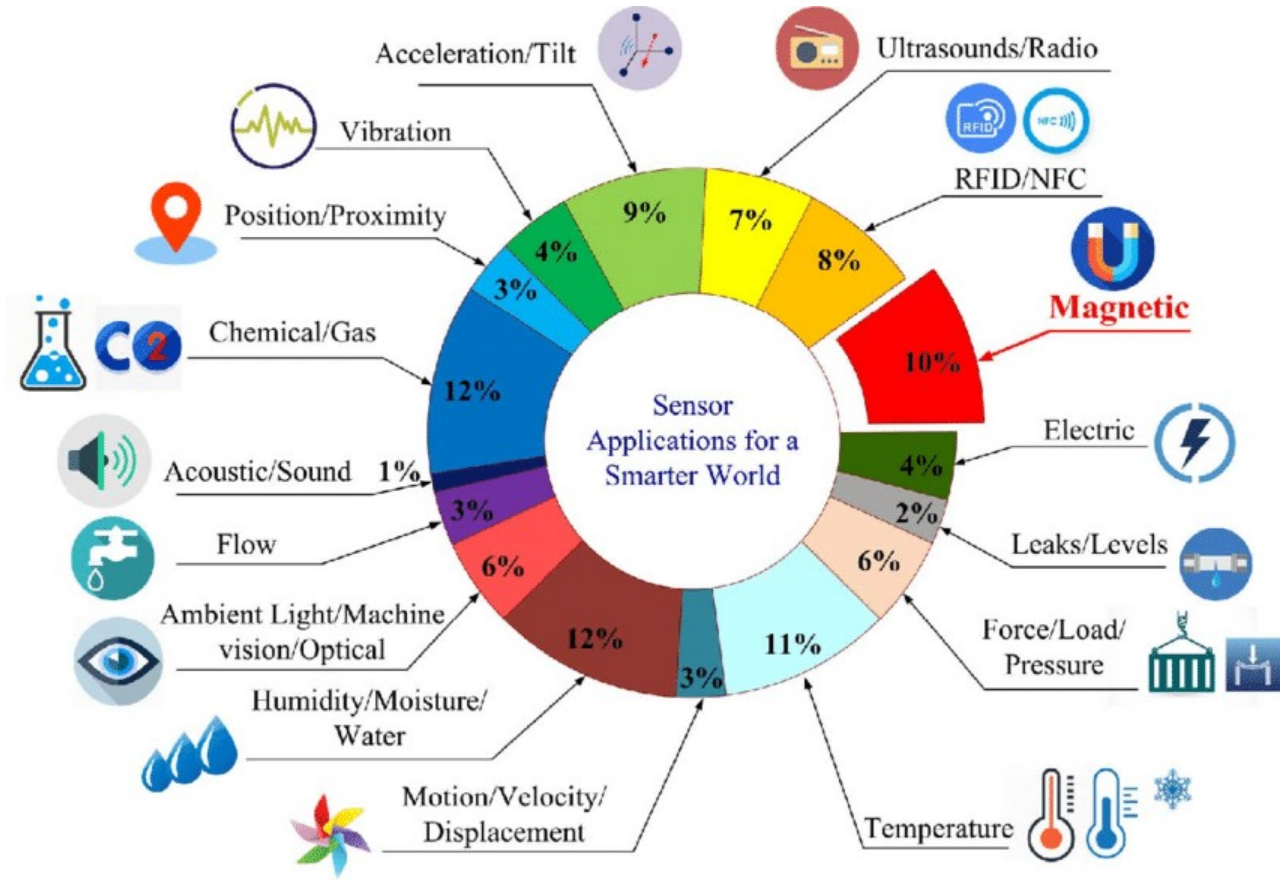
How IoT Works



Sensors and devices talk to the cloud; software processes the data and acts, or shows it to a user.

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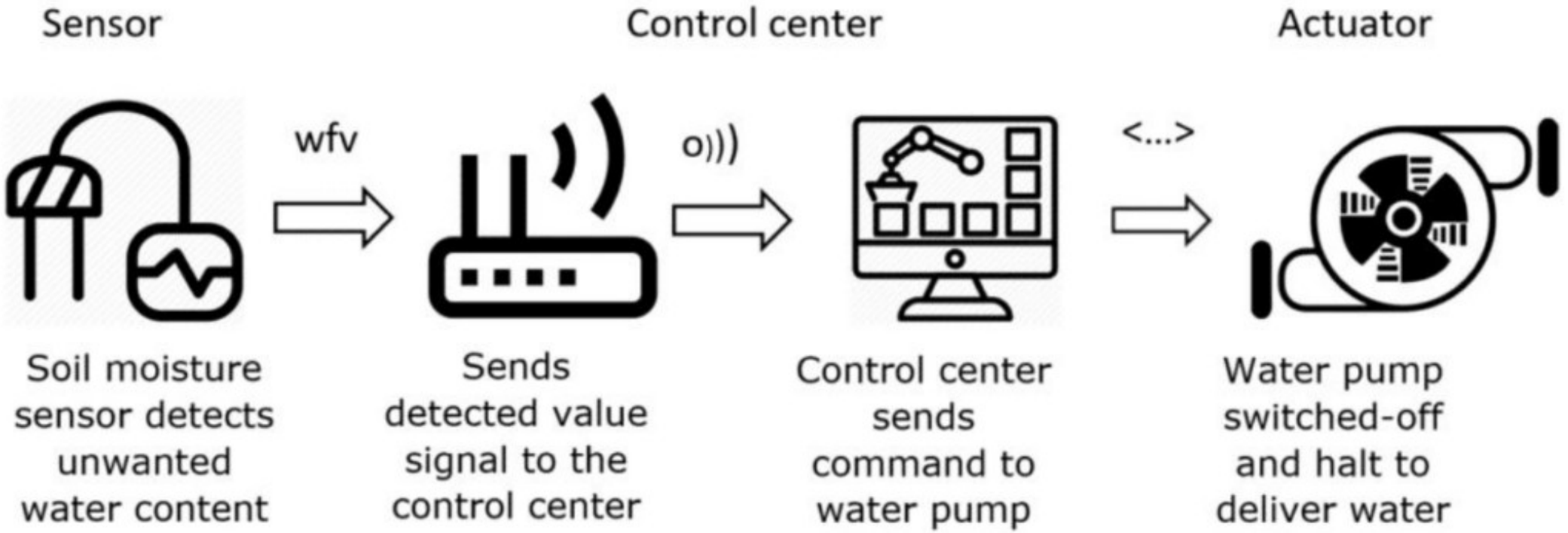
Sensors for IoT



Low-cost sensors are what made large-scale IoT deployment economically viable.

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Actuators for IoT



Actuators turn a decision in the cloud back into a physical action.

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Sensors, Actuators and Gateways

Sensors — sense

- Temperature & humidity
- Motion, presence, vibration
- Light, gas, air quality
- GPS location, current

Actuators — act

- Relays and switches
- Motors and servos
- Valves and pumps
- Alarms and displays

Gateways — connect

- Aggregate nearby devices
- Translate protocols
- Buffer when offline
- Add edge processing

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CONNECTIVITY CHOICES

Wireless Technologies for IoT

1

WiFi

High bandwidth, short range, mains power — best indoors where power is available.

2

Bluetooth / BLE

Very low power, very short range — wearables and personal devices.

3

Cellular (4G/5G)

Wide coverage with a subscription cost — vehicles, logistics and remote assets.

4

LoRaWAN / NB-IoT

Long range, very low power, tiny payloads — agriculture, metering, city-scale sensing.

5

Zigbee / Z-Wave

Low-power mesh — smart buildings and home automation.

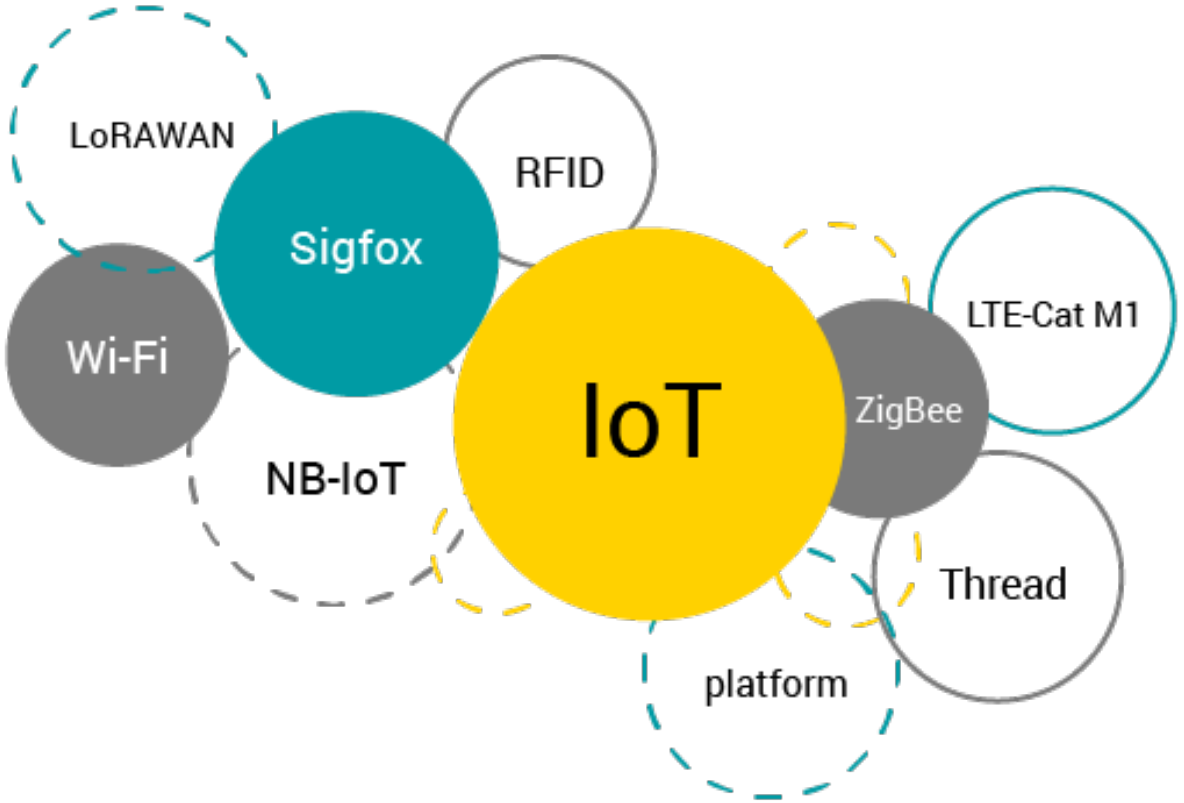
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Choosing one

Trade range against power, bandwidth and cost — there is no single best option.

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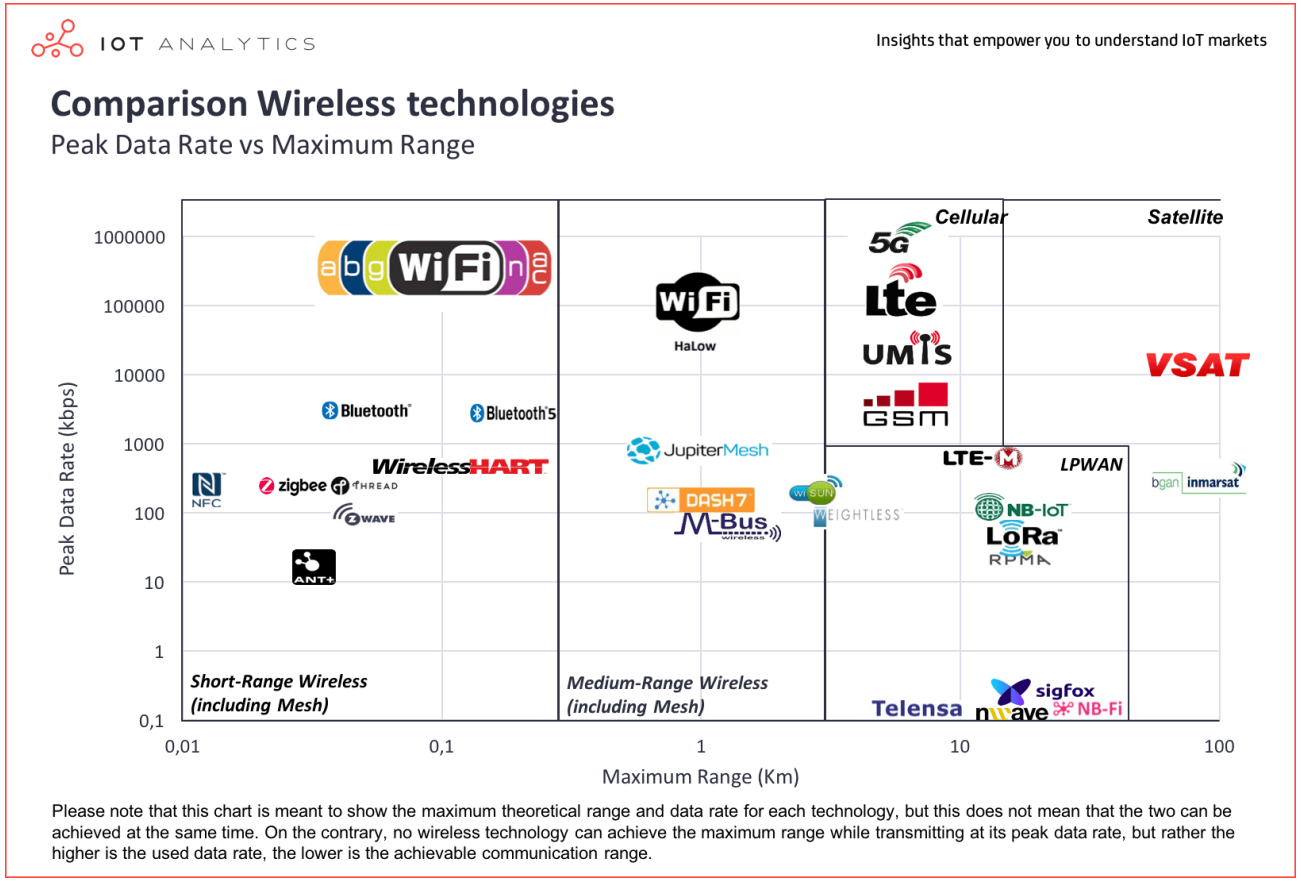
Wireless Technologies for IoT



Range, power consumption, bandwidth and cost determine which technology fits a use case.

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Comparing Wireless Technologies



There is no single best option — every choice trades range against power and bandwidth.

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New Business Models Enabled by IoT

1

Product → Service

Sell the outcome (cooling, uptime, mobility) rather than the machine itself.

2

Subscription

Recurring revenue for monitoring, analytics and support built on the data stream.

3

Pay-per-use

Charge by actual consumption, measured by the device — fair for the customer, predictable for you.

4

Predictive maintenance

Service before failure; convert unplanned downtime into scheduled work.

5

Data-driven products

The telemetry itself becomes a product — benchmarking, insurance, optimisation.

6

Ecosystem platforms

Open the data to partners and earn from the platform, not just the device.

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TOPIC 01 · ACTIVITY

Discussion — Compare Business Models

DISCUSSION · 15 MIN

In pairs, describe how your organisation earns revenue today. Then pick one IoT-enabled competitor or industry example and compare the two.

1. How does your organisation make money today — product sale, project fee, subscription?
2. Which IoT-enabled model could apply: outcome-based, pay-per-use, subscription, or predictive service?
3. What data would you need to collect before that model becomes possible?
4. What would customers gain, and what would they be willing to pay for?

YOUR OUTPUT A short written comparison of your current model against one IoT-enabled model.

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TOPIC RECAP

Recap — Overview of Internet of Things (IoT)

- IoT connects physical assets to the internet so they report their own state.
- Four building blocks: sensors/devices, connectivity, cloud, user interface.
- Falling sensor cost, low-power wireless and 5G made mass adoption viable.
- IoT shifts firms from product sales to service, subscription and outcome models.

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02

TOPIC 02

Potential Opportunities of IoT Innovation

Agriculture · Autonomous vehicles · Logistics · Smart buildings · Healthcare · Drones

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THE OPPORTUNITY LANDSCAPE

IoT has reached every industry.

From agriculture to aerospace — the pattern is always the same: sense something that used to be invisible, then act on it.

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LO2 · A2, K1 · 60 min

Key Concepts — Potential Opportunities of IoT Innovation

1

Agriculture

Soil, weather and crop sensing drive precision irrigation and fertiliser decisions — e.g. IBM AgroPad.

2

Autonomous vehicles

Vehicle telemetry and sensor fusion enable driver assistance and highly automated driving.

3

Logistics & fleet

Location, temperature and fuel telemetry give end-to-end shipment visibility and route optimisation.

4

Smart buildings

Occupancy, energy and equipment sensing cut utility cost and enable predictive maintenance.

5

Healthcare & wearables

Continuous patient monitoring feeds records securely, supporting earlier clinical intervention.

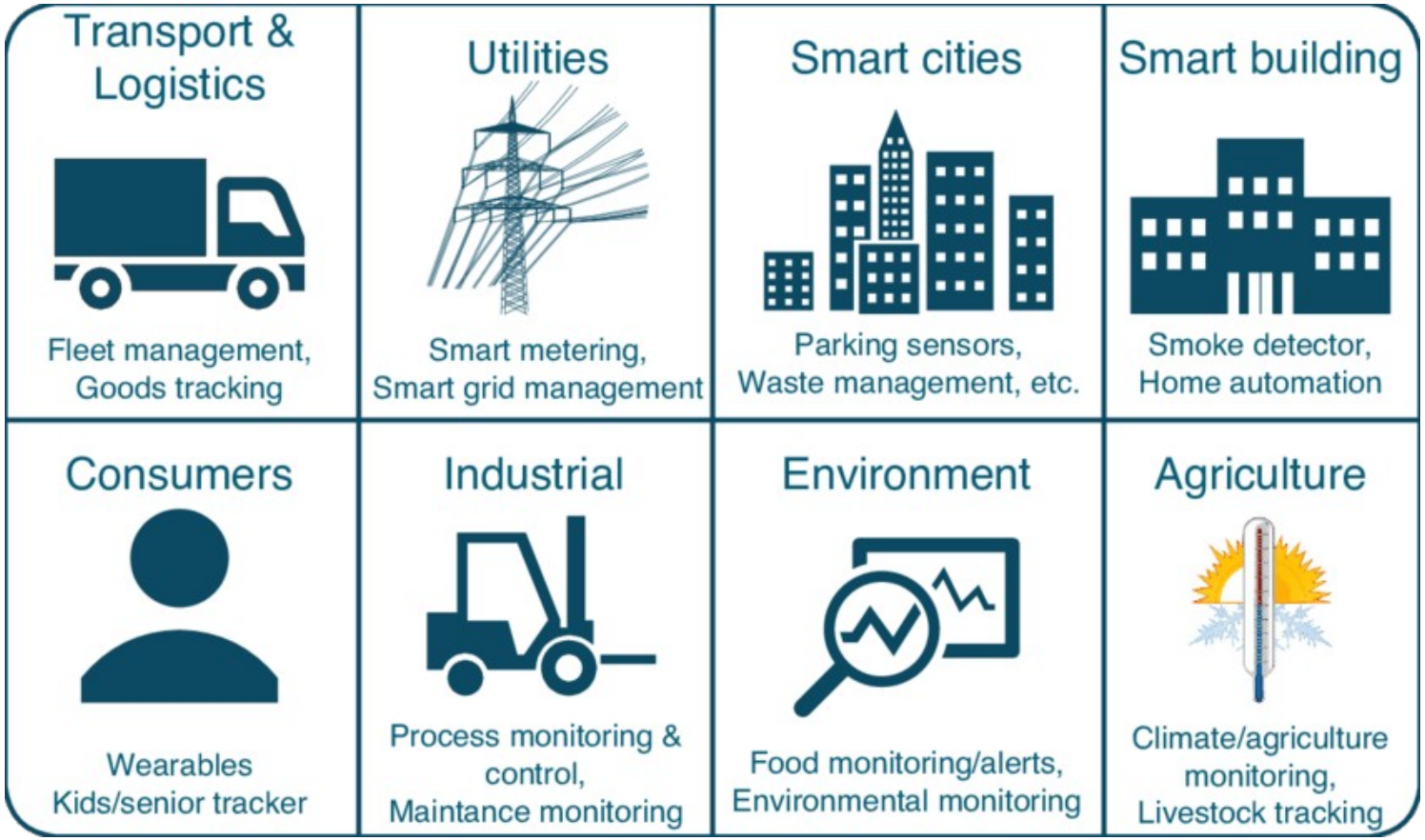
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Drones

Aerial inspection, survey and delivery reach places people cannot, safely and at lower cost.

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IoT Applications Across Industry



IoT has found its way into every conceivable field, from agriculture to aerospace.

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IoT for Agriculture



Precision agriculture: sense the soil and crop, then irrigate and fertilise to the actual need.

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Use Case — Agriculture

The problem

- Soil and crop condition is invisible
- Water and fertiliser applied uniformly
- Decisions made from experience alone

The IoT solution

- Soil moisture & nutrient sensing
- Weather and micro-climate data
- IBM AgroPad: instant chemical test

The business value

- Precision irrigation cuts water use
- Higher yield per hectare
- Lower input cost per tonne

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Use Case — Autonomous Vehicles & Logistics

Autonomous driving

- Sensor fusion: radar, lidar, camera
- Driver assistance to full automation
- BMW: 30 years of development

Fleet monitoring

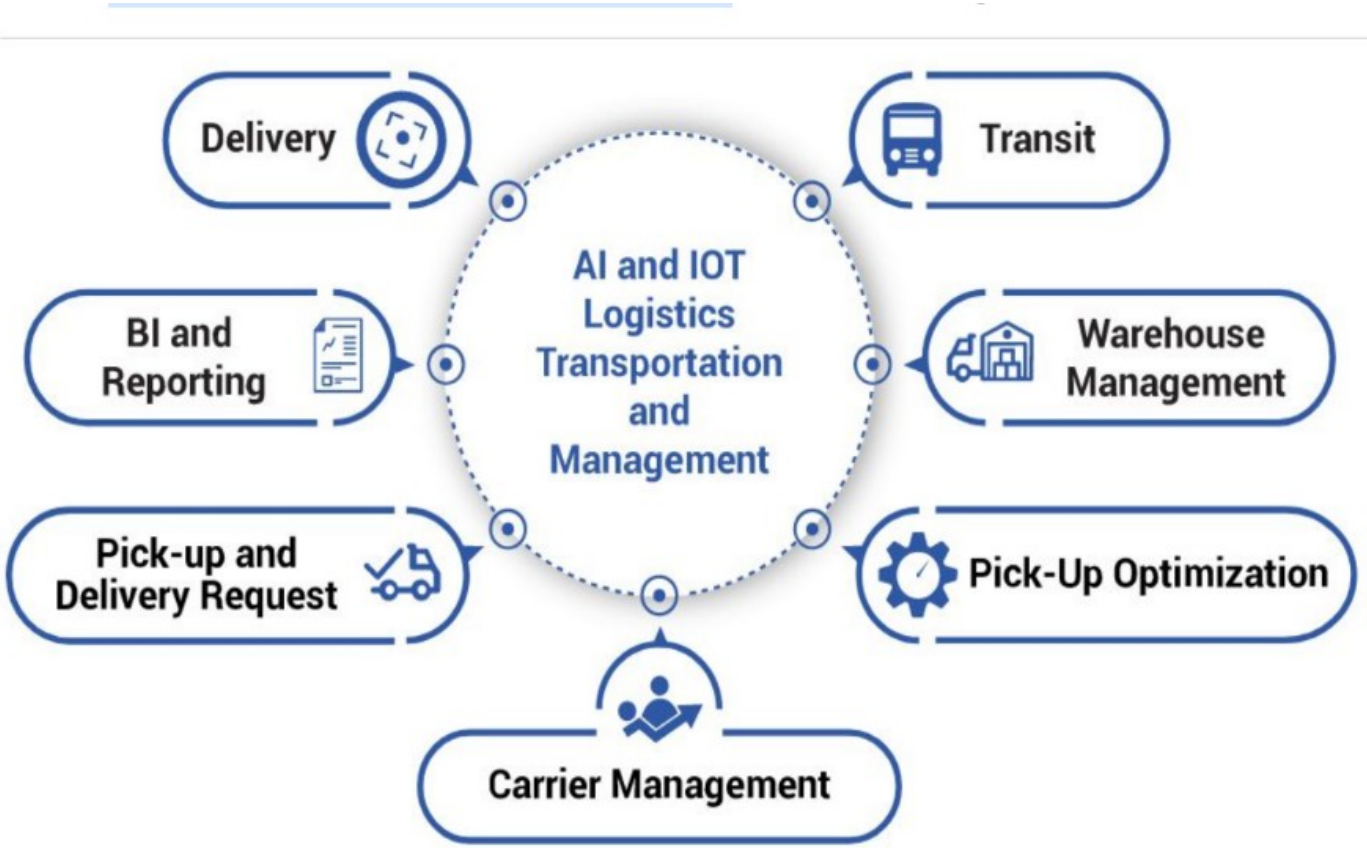
- Location, fuel and engine telemetry
- Cold-chain temperature tracking
- Oracle IoT Fleet Monitoring

The business value

- Route and fuel optimisation
- Fewer accidents and claims
- End-to-end shipment visibility

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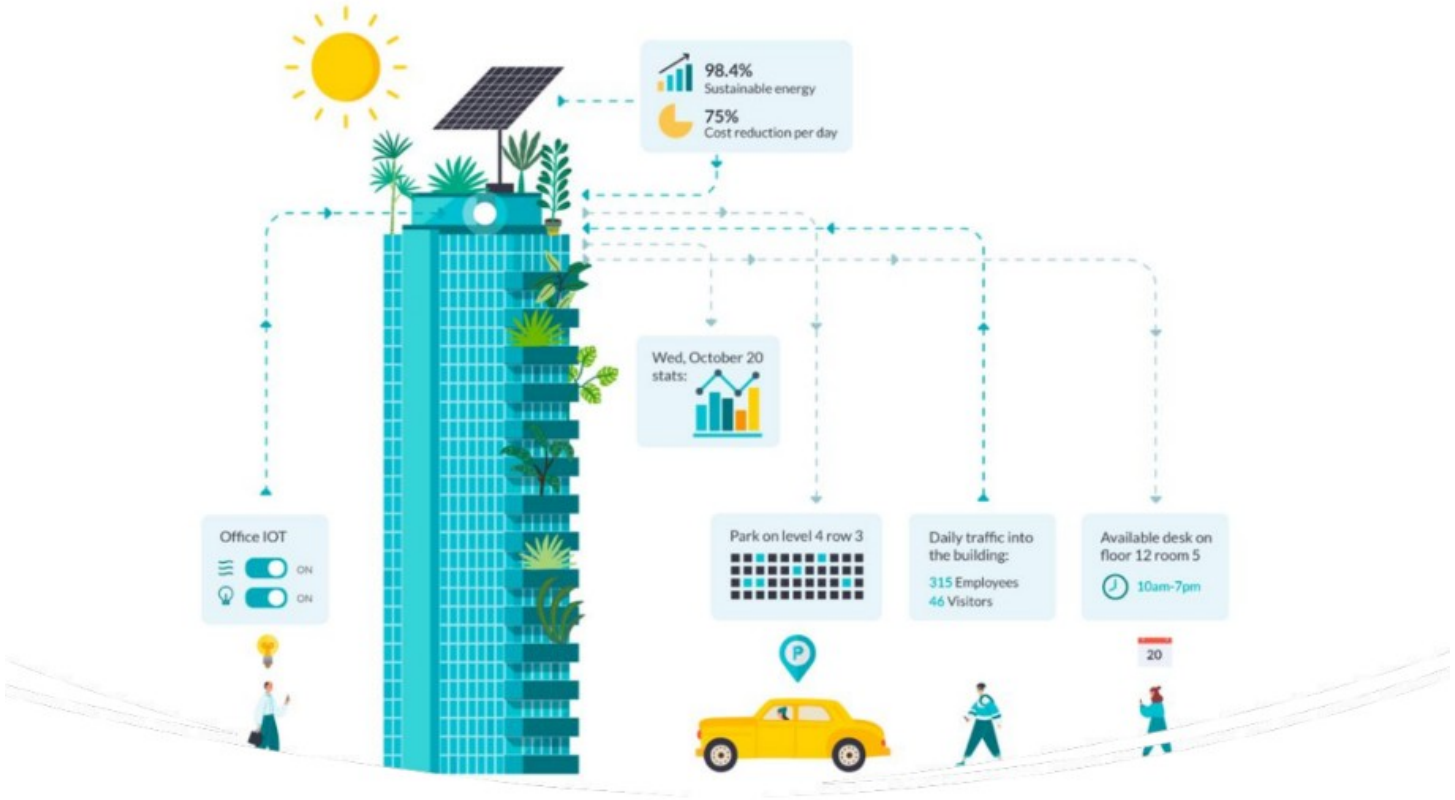
IoT for Logistics and Fleet



Location, temperature and fuel telemetry give end-to-end shipment visibility.

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IoT for Smart Buildings



Occupancy, energy and equipment sensing cut utility cost and enable predictive maintenance.

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IoT for Healthcare



Continuous patient monitoring feeds the medical record and supports earlier intervention.

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Use Case — Buildings, Healthcare & Drones

Smart buildings

- Occupancy and energy sensing
- Predictive lift & HVAC maintenance
- Smart washroom servicing on demand

Healthcare & wearables

- Continuous patient monitoring
- Philips: data feeds the EMR securely
- Earlier clinical intervention

Drones

- Aerial inspection and survey
- Singapore offshore ship delivery
- Reaches where people cannot safely go

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How to Find an IoT Opportunity

1. Find a process that is invisible, manual or reactive today
2. Decide what to MEASURE and how often
3. Decide who or what ACTS on that reading
4. Quantify the cost of the current blind spot
5. Pilot on one asset before you scale

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TOPIC 02 · ACTIVITY

Discussion — Spot an IoT Opportunity

DISCUSSION · 15 MIN

Working from the six industry use cases, identify one process in your own organisation that is currently invisible, manual or reactive, and could be sensed instead.

1. Which process today relies on someone physically checking something?
2. What would you measure, how often, and who would act on the reading?
3. What is the cost of the current blind spot — downtime, spoilage, overtime, penalties?
4. Which of the six use cases is closest to your scenario, and what can you borrow from it?

YOUR OUTPUT One clearly stated IoT opportunity: the asset, the measurement, and the decision it improves.

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TOPIC RECAP

Recap — Potential Opportunities of IoT Innovation

- IoT opportunities exist in every sector — agriculture, mobility, logistics, buildings, health and aerial.
- The repeating pattern: sense what was invisible, then act on it automatically.
- Borrow the pattern from an adjacent industry rather than inventing from scratch.
- Start from the business pain and the cost of the blind spot, not from the technology.

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03

TOPIC 03

IoT Technology

Cloud platforms · Post data · Read & control · Analytics & visualisation · Automation

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LO3 · A3, K2 · 60 min + 90 min practical

Key Concepts — IoT Technology

1

Cloud for IoT

A platform ingests telemetry, stores history, visualises it and triggers actions — no servers to run.

2

IoTFlow

Our platform: projects, devices, tokens, dashboards, alerts and automations in one place.

3

Uplink: posting data

Devices POST readings over HTTP REST or publish over MQTT, authenticated by a device token.

4

Downlink: control

Virtual pins carry commands from the dashboard back to the device to switch a relay or set a value.

5

Analytics & visualisation

Number cards, gauges, charts and maps turn raw readings into an operational picture.

6

Automation & AI

Rules and n8n workflows notify, log and let AI summarise readings and recommend an action.

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Cloud Computing for IoT

1

Why the cloud

Devices are small and intermittent; the cloud gives storage, compute and availability they cannot.

2

What a platform does

Ingest telemetry, store history, visualise it, raise alerts and trigger actions.

3

No servers to run

A managed IoT platform removes the infrastructure work so you focus on the business logic.

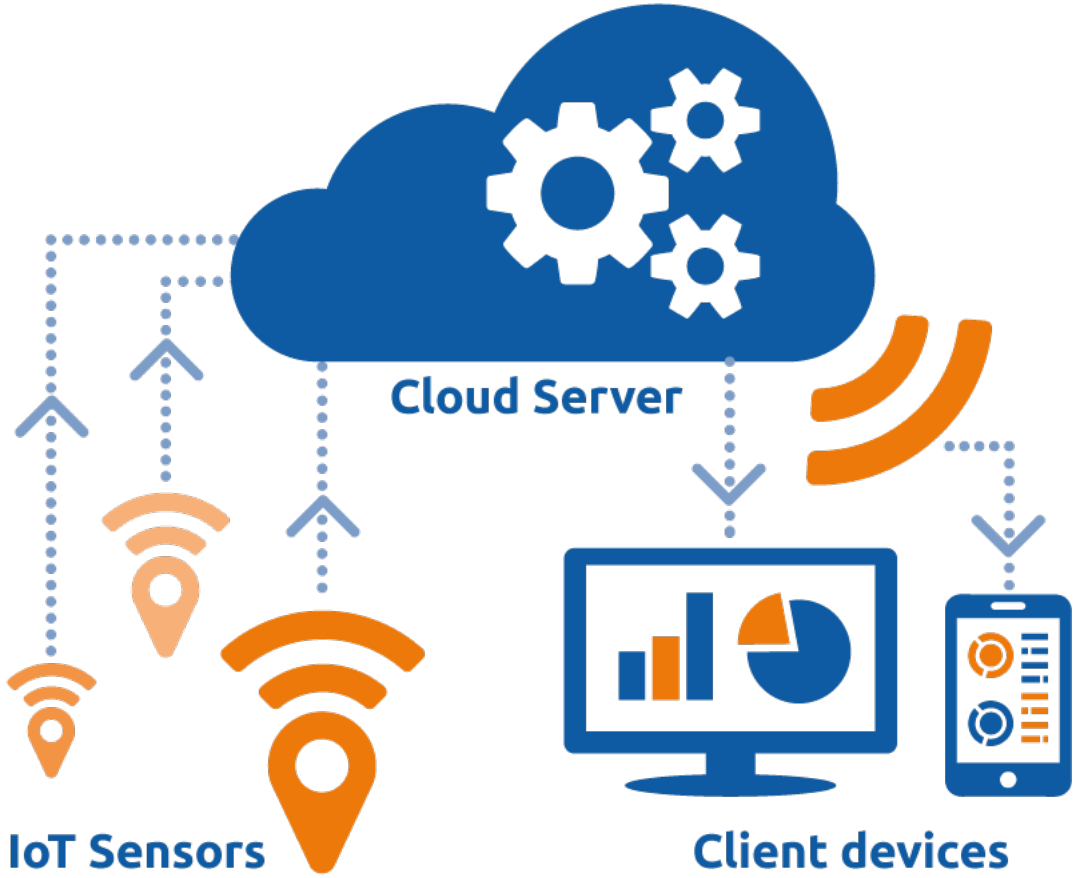
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Scales with you

One device or ten thousand — the same architecture, priced by what you actually use.

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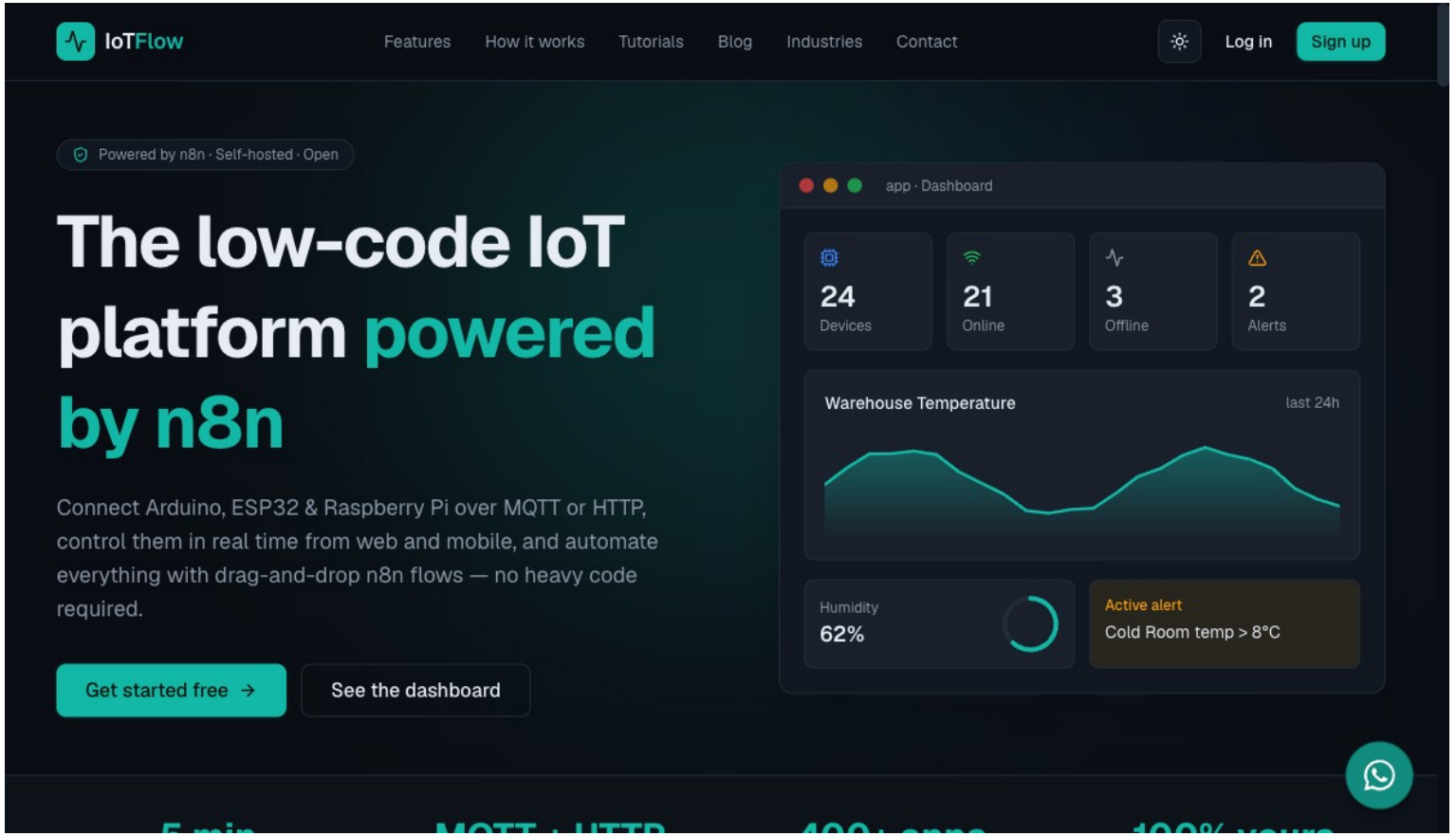
Cloud Computing for IoT



The cloud gives constrained devices the storage, compute and availability they lack.

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IoTFlow — https://iot.tertiaryinfotech.com



The platform you will use for every lab in this course.

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https://iot.tertiaryinfotech.com

IoTFlow — Our IoT Platform

1

Projects

A self-contained workspace with its own devices, dashboard, alerts, map and automations.

2

Devices & tokens

Register a device in a guided wizard; a device token authenticates every message it sends.

3

HTTP & MQTT

POST JSON to a REST endpoint, or publish/subscribe over MQTT for real-time two-way traffic.

4

Dashboards

Number cards, gauges, charts, LED indicators and maps that refresh automatically.

5

Alerts & automations

Threshold and offline rules that fire a webhook into an n8n workflow.

6

Clients

Official Python client (pip install iotflow) and an Arduino library for ESP32/ESP8266.

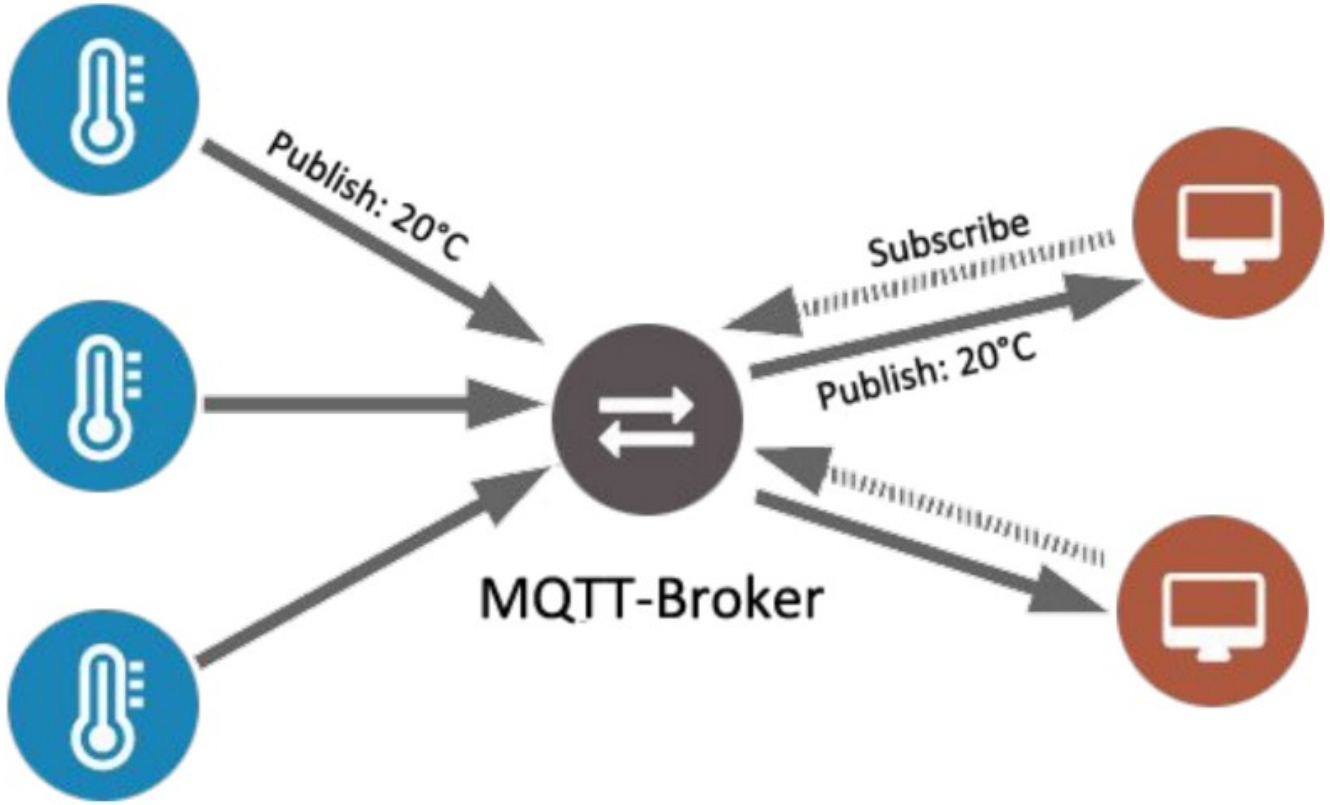
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The IoT Data Round Trip

1. DEVICE reads a sensor value
2. UPLINK posts it over HTTP or MQTT
3. CLOUD stores, charts and evaluates rules
4. DECISION — a person or an automation acts
5. DOWNLINK sends a command back to the device

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How MQTT Works



Publish/subscribe: devices publish to a topic and the broker delivers to every subscriber.

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TWO DIRECTIONS, TWO PURPOSES

Uplink vs Downlink

Monitoring

- Uplink — telemetry
  - Device → cloud
  - Sensor readings and status
  - HTTP POST /api/telemetry
  - or MQTT publish
  - This is your monitoring data

Control

- Downlink — commands
  - Cloud → device
  - Switch a relay, set a value
  - MQTT topic devices/<id>/down
  - or HTTP state polling
  - This is your automation

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Turning Data into Business Insight

1

Visualise

Number cards, gauges, line charts and maps make current performance obvious at a glance.

2

Detect

Alert rules fire the moment a reading crosses a business-meaningful threshold.

3

Notify

Automations push the event to email, Telegram or WhatsApp so the right person knows.

4

Record

Append every event to a sheet or database to build the history the business case needs.

5

Analyse with AI

An AI node summarises recent readings and recommends an action in plain language.

6

Act automatically

Close the loop — send a command back to the device with no human in the middle.

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TOPIC 03 · PRACTICAL

Hands-On Practical — 90 minutes

Four labs on IoTFlow (https://iot.tertiaryinfotech.com). Step-by-step instructions are in your Learner Guide — follow along there.

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WHAT YOU'LL DO

Hands-On Labs — IoT Technology

Labs 1–2

- Register your first IoT device
- Post business data to the cloud

Lab 3

- Read data back
- Remotely control the asset

Lab 4

- Dashboard, alerts & AI automation
- Turn data into a decision

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TOPIC 03 · HANDS-ON

Register Your First IoT Device on IoTFlow

LAB 1

Identify how digitisation is applied to a business asset by connecting it to the cloud (A3, K2).

Turn a physical asset into a digital one. You sign in to IoTFlow, create a project workspace for your business scenario, register a device through the Add Device wizard, choose how it will talk to the cloud (HTTP or MQTT), and capture the device token that authenticates every message it sends.

YOU'LL BUILD

A registered IoT device in your own IoTFlow project, with its device token saved

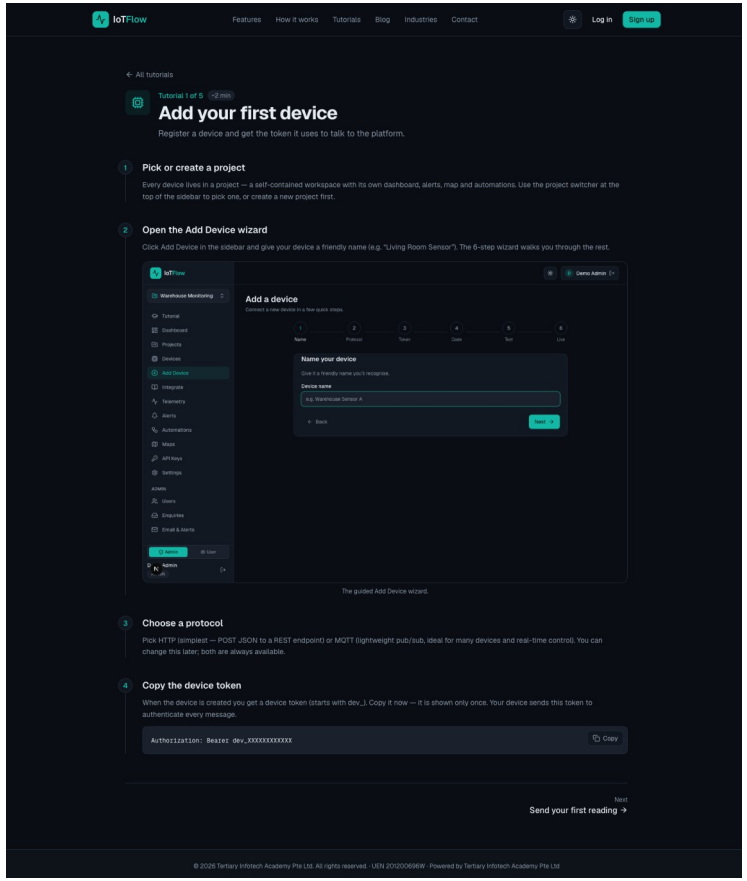
WHY THE BUSINESS CARES

This is the first step of digitisation: an asset that was previously invisible to the business now has an identity, an owner and a data stream that can be measured and managed.

TOOLS IoTFlow projects · Add Device wizard · Device tokens · HTTP & MQTT

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Lab 1 — Register Your First IoT Device on IoTFlow



IoTFlow — A registered IoT device in your own IoTFlow project, with its device token saved

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TOPIC 03 · HANDS-ON

Post Business Data to the Cloud (HTTP & MQTT)

LAB 2

Apply digitisation by streaming operational data from the asset to the cloud (A3, K2).

Send readings to IoTFlow three different ways — cURL from any terminal, the Python client, and (optionally) an ESP32/ESP8266 over MQTT — and watch them arrive live. This is the uplink that every IoT business case depends on: without a reliable data stream there is nothing to analyse.

YOU'LL BUILD

Live temperature and humidity telemetry streaming into IoTFlow from terminal and Python

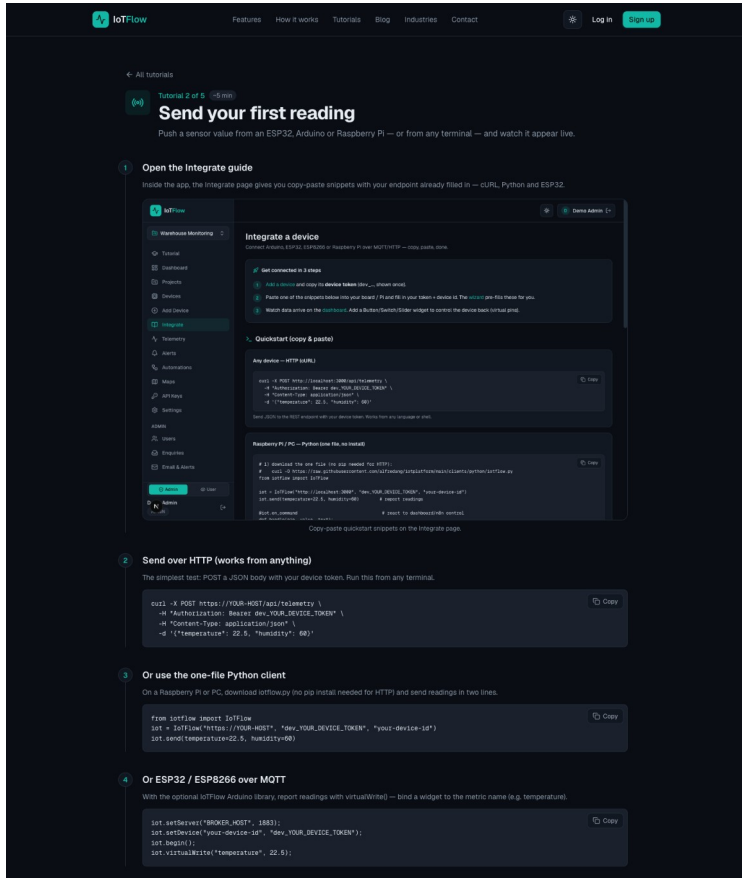
WHY THE BUSINESS CARES

Each reading is an operational fact with a timestamp. Once the stream exists, the business can ask questions it previously could not answer — how often, how long, how bad, and at what cost.

TOOLS Telemetry REST API · Python client (pip install iotflow) · MQTT broker · ESP32/ESP8266

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Lab 2 — Post Business Data to the Cloud (HTTP & MQTT)



IoTFlow — Live temperature and humidity telemetry streaming into IoTFlow from terminal and Python

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TOPIC 03 · HANDS-ON

Read Data Back and Remotely Control the Asset

LAB 3

Close the loop between cloud data and physical action — the basis of an IoT business process (A3, K2).

Read device state back out of the cloud over the REST API and MQTT, then add Control widgets (switch, slider) bound to virtual pins and drive the device from the dashboard. Reading data is monitoring; sending commands back is automation — and that is where the labour savings appear.

YOU'LL BUILD

A dashboard switch and slider that control a relay/LED on the device in real time

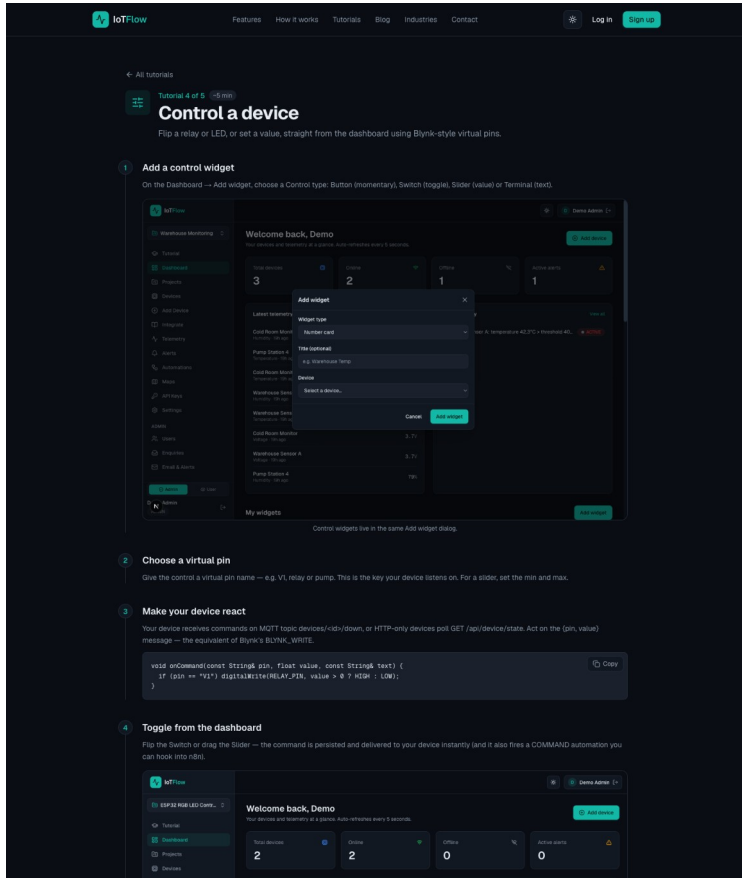
WHY THE BUSINESS CARES

Monitoring tells you what happened. Control lets the system act without a person driving out to the site — this is usually the single largest line in an IoT cost-benefit case.

TOOLS REST API (GET /api/device/state) · MQTT subscribe · Control widgets · Virtual pins · @on_command

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Lab 3 — Read Data Back and Remotely Control the Asset



IoTFlow — A dashboard switch and slider that control a relay/LED on the device in real time

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TOPIC 03 · HANDS-ON

Turn IoT Data into Business Insight

LAB 4

Use analytics, visualisation and automation to turn raw telemetry into a business decision (A3, K2).

Build a live dashboard from your readings — number cards, gauges, charts, LED indicators and maps — then set a threshold alert and hand it to an n8n workflow that notifies the team, logs the event and asks an AI node to summarise the readings and recommend an action.

YOU'LL BUILD

A monitoring dashboard plus an alert-driven n8n workflow with an AI summary and a downlink command

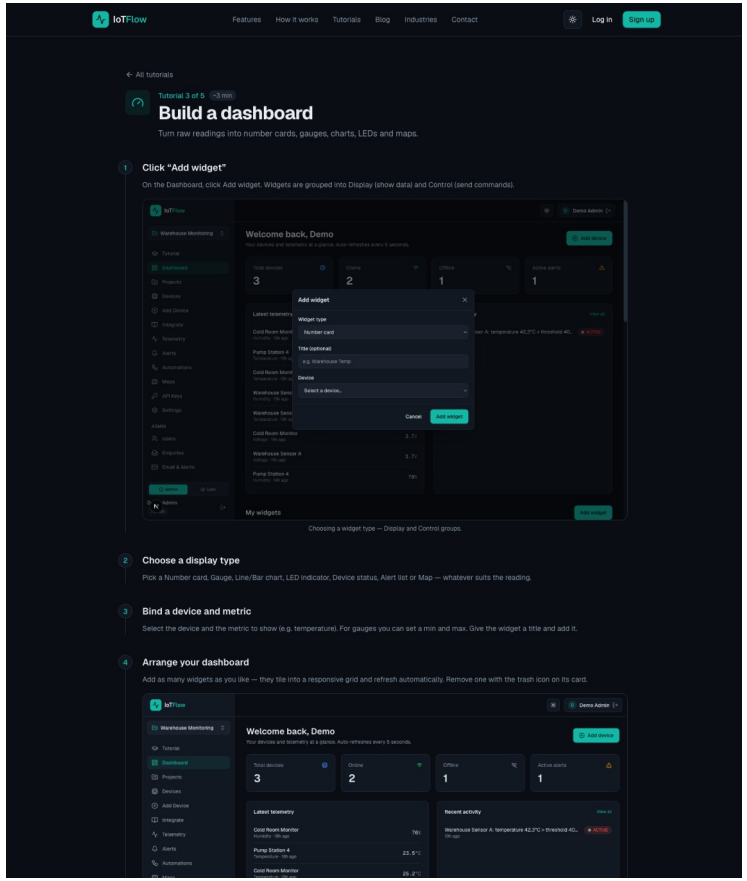
WHY THE BUSINESS CARES

This is the slide the board actually sees. A dashboard makes performance visible, an alert makes it timely, and the AI summary turns a stream of numbers into a recommended action.

TOOLS Display widgets · Alert rules · IoTFlow Automations · n8n Webhook · Email/Telegram · Google Sheets · AI nodes

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Lab 4 — Turn IoT Data into Business Insight



IoTFlow — A monitoring dashboard plus an alert-driven n8n workflow with an AI summary and a downlink command

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What the Labs Proved

1

Any asset can be digitised

A token and a few lines of code put a physical thing on the network.

2

Data arrives in seconds

The dashboard updated live as each reading was posted.

3

Control closes the loop

A command from the dashboard changed the device state remotely.

4

Insight can be automatic

A threshold fired a workflow that notified, logged and recommended an action.

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TOPIC RECAP

Recap — IoT Technology

- An IoT platform ingests telemetry, stores it, visualises it and triggers actions — with no servers to run.
- Uplink carries sensor readings to the cloud; downlink carries commands back to the device.
- Dashboards make performance visible; alert rules make problems timely.
- Automations and AI turn a stream of readings into a recommended business action.

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04

TOPIC 04

IoT Adoption

Adoption challenges · IoT security · Edge computing · Cost-benefit analysis

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LO4 · A4, K3 · 60 min

Key Concepts — IoT Adoption

1

Barriers to adoption

Data privacy, security, cost, skills gaps, weak standards and interoperability with legacy systems.

2

Why IoT security differs

Constrained devices, long unattended lifetimes and physical consequences to a breach.

3

Mirai botnet

Default credentials on cameras and routers built a botnet that took down much of the US east coast.

4

Edge computing

Processing near the source cuts latency, bandwidth and cost, and keeps sensitive data local.

5

Falling hardware cost

Sensor, gateway and connectivity prices keep dropping, improving the business case each year.

6

Cost-benefit analysis

Weigh hardware, connectivity, platform and integration against savings, uptime and new revenue.

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TOP BARRIERS TO IoT ADOPTION

Why Organisations Hold Back

1

Data privacy

Who owns the data, where does it live, and what does the law require?

2

Security

Connected devices widen the attack surface into the physical world.

3

Cost of implementation

Hardware, integration and change management, not just the sensors.

4

Knowledge gap

Teams do not know which solutions exist or how to evaluate them.

5

Legacy & interoperability

Existing systems do not speak the same protocols.

6

Unproven benefit

Uncertainty that IoT will actually deliver what was promised.

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IoT Security — Why It Is Different

Constrained devices

- Microcontrollers, not servers
- Often battery powered
- May not support TLS/SSL

Unattended & long-lived

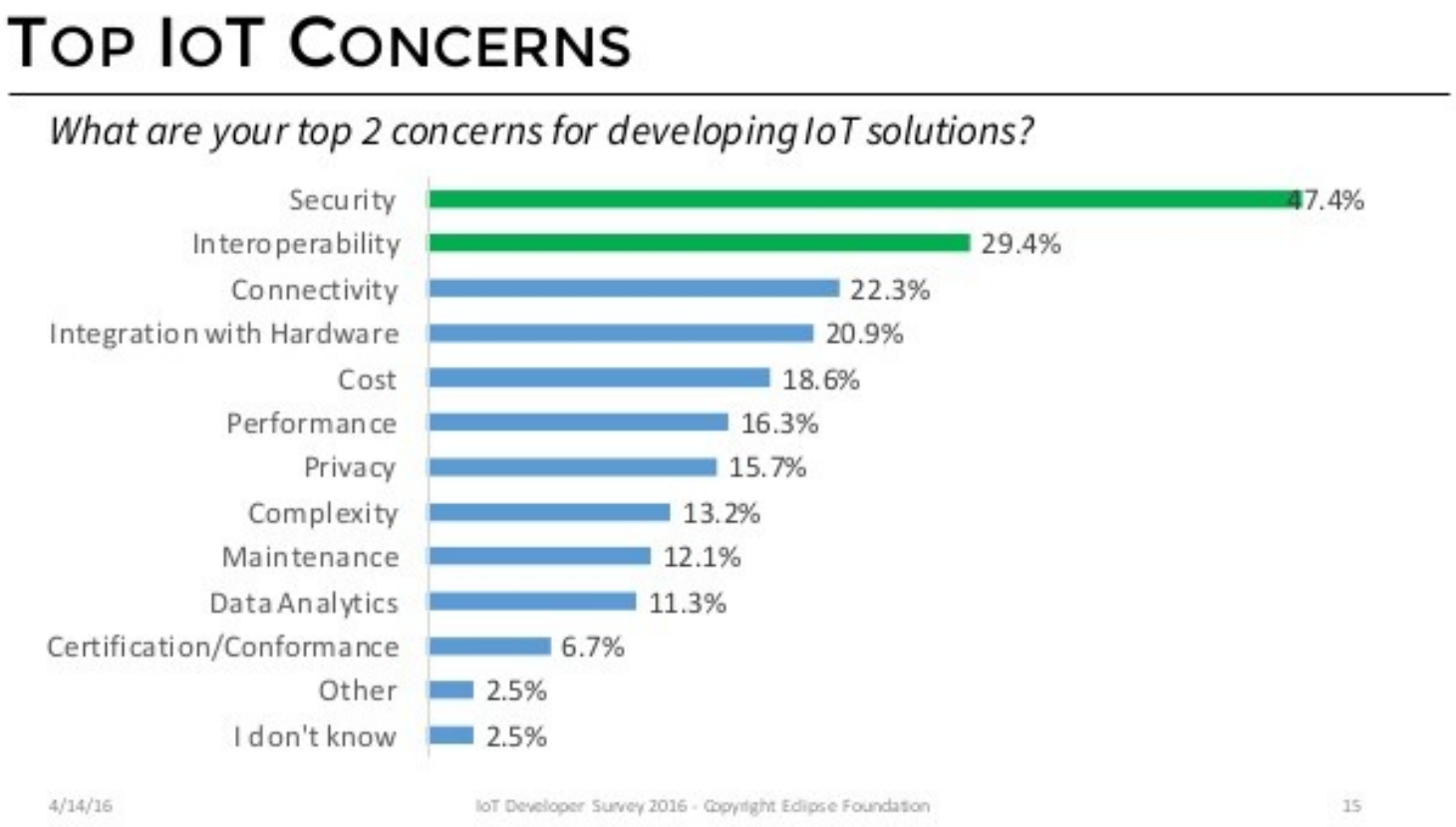
- Run for years without supervision
- Patching is inconvenient
- Rogue devices go unnoticed

Physical consequences

- IT breach leaks data
- IoT breach can cause harm
- Safety-critical systems at risk

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IoT Security Is a Top Concern



Security and data privacy are consistently the biggest barriers to IoT adoption.

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CASE STUDY · WHY DEVICE SECURITY MATTERS

The Mirai botnet, 12 October 2016.

Default credentials on cameras and routers were harvested into a botnet whose DDoS attack took much of the US east coast offline — an unsecured device is everyone's problem.

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THE SOLUTION STACK

The Four Stages of an IoT Solution

1

SENSE — devices and sensors capture the physical state

2

CONNECT — gateways and networks carry the data

3

PROCESS — cloud or edge analyses and stores it

4

ACT — dashboards, alerts and automations drive the decision

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Edge vs Cloud Processing

Near the data

- Edge computing
  - Process at or near the device
  - Lower latency — acts immediately
  - Less bandwidth and lower cost
  - Sensitive data stays local
  - Limited compute and storage

Central

- Cloud computing
  - Process in the data centre
  - Unlimited compute and history
  - Easier to update and scale
  - Best for analytics and AI
  - Needs connectivity and bandwidth

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Cost-Benefit Analysis of IoT

1

Hardware

Sensors, actuators, gateways — prices have fallen steadily year on year.

2

Connectivity

SIM, LoRa or WiFi; recurring per-device cost that scales with the fleet.

3

Platform

Subscription by device count or data volume; low-budget options exist.

4

Integration

Often the largest line — connecting IoT data into existing business systems.

5

Benefits

Labour saved, downtime avoided, energy and wastage reduced, new revenue enabled.

6

Payback

Compare total cost against annual benefit; pilot first to validate the assumptions.

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TOPIC 04 · ACTIVITY

Discussion — Cost-Benefit Analysis

DISCUSSION · 20 MIN

Take the opportunity you identified in Topic 2 and build a first-pass feasibility and cost-benefit case for it, including the security and privacy risks.

1. Costs: devices, connectivity, platform subscription, integration, installation, maintenance.
2. Benefits: labour saved, downtime avoided, energy or wastage reduced, new revenue enabled.
3. Risks: data privacy, device security, interoperability with legacy systems, skills gaps.
4. Roughly how long before the benefits exceed the costs — and what would break that estimate?

YOUR OUTPUT A cost-benefit summary with an estimated payback period and the top three risks.

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TOPIC RECAP

Recap — IoT Adoption

- Privacy, security, cost, skills and legacy interoperability are the main adoption barriers.
- IoT security differs from IT security: constrained, unattended devices with physical consequences.
- Edge processing cuts latency, bandwidth and privacy exposure; the cloud gives scale and analytics.
- A credible cost-benefit case counts integration and running cost, not just the hardware.

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05

TOPIC 05

IoT Implementation

Success factors · Design thinking · A 7-step action plan

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LO5 · A5, K4 · 60 min

Key Concepts — IoT Implementation

1

Lead with business value

Start from the business outcome IoT delivers, not from the technology itself.

2

Strategic alignment

The initiative must serve the business strategy and respect technology constraints.

3

Business process focus

Integrate IoT into existing workflows so the new way is clearly better than today.

4

Operating model & capability

Value networks and new skills are needed; give people a common IoT language.

5

Security by design

Build end-to-end security in from the start — it is a selection criterion, not an add-on.

6

Design thinking

Empathise, Define, Ideate, Prototype, Test — start small, prove value, then scale.

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Six Success Factors for IoT Implementation

1. **Business benefits**
   Focus on the value IoT delivers, not on what the technology can do.

2. **Strategic alignment**
   Align with the business strategy and respect technology constraints.

3. **Business process focus**
   Integrate into existing workflows so the new way is clearly better.

4. **Operating model changes**
   Build the value network — end-to-end IoT needs close collaboration.

5. **Capability uplift**
   Teach people IoT and Industry 4.0 so everyone shares a common language.

6. **End-to-end security**
   Security by design becomes a criterion when choosing technology.

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Design Thinking — the Stanford d.school Model

1. EMPATHISE — understand the user's real experience
2. DEFINE — state the problem from their point of view
3. IDEATE — generate many diverse possible solutions
4. PROTOTYPE — make it tangible enough to interact with
5. TEST — observe, gather feedback and refine

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THE 7-STEP ACTION PLAN

Applying Design Thinking to Your IoT Project

1

1 · Set business focus

Commit at top-management level and state the strategic goal IoT must serve.

2

2 · Introduce IoT capability

Build the team's understanding of what IoT can realistically do.

3

3 · Empathise with the user

Identify the pains and gains of the customer or operator affected.

4

4 · Ideate and select

Generate options, then select on brand values and business ambition.

5

5 · Prototype a pilot

Smallest deployment that proves value — one asset, one site, one metric.

6

6 · Identify implications

Develop and validate the business case, including risk and security.

7

7 · Pitch to decision makers

Win the coalition and the funding needed to scale beyond the pilot.

8

Then scale

Roll out on evidence from the pilot, not on optimism.

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TOPIC 05 · ACTIVITY

Discussion — Your Implementation Action Plan

DISCUSSION · 20 MIN

Apply design thinking and the success factors to turn your opportunity into a staged action plan you could actually pitch to management.

1. Empathise & Define: whose problem is this, and how would you state it in one sentence?
2. Ideate & Prototype: what is the smallest pilot that proves the value — one device, one site?
3. Test & scale: what measurable result would justify rolling it out further?
4. Which success factor is weakest in your organisation today, and how will you address it?

YOUR OUTPUT A staged action plan: pilot scope, success measure, owner, and the case for scaling.

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TOPIC RECAP

Recap — IoT Implementation

- Lead with business value and align the initiative with the business strategy.
- Integrate IoT into existing processes; build capability and the value network around it.
- Design security in from the start — it is a selection criterion, not an afterthought.
- Use design thinking: empathise, define, ideate, prototype, test — pilot before you scale.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved Today

1

LO1 · Business models

Compared your current business model against IoT-enabled models.

2

LO2 · Opportunities

Explored IoT use cases across six industries and found one of your own.

3

LO3 · How IoT works

Registered a device, posted data, controlled it remotely and built a dashboard.

4

LO4 · Costs & benefits

Weighed adoption barriers, security risk and the cost-benefit case.

5

LO5 · Implementation

Applied design thinking and the success factors to a staged action plan.

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AFTER TODAY

Your Next Steps

1

Keep your IoTFlow project

Your devices, dashboard and automations stay available after the course.

2

Run the pilot you designed

Start with one asset and one metric — prove the value before scaling.

3

Revisit the Learner Guide

Every lab's step-by-step instructions are there for you to repeat.

4

Explore the tutorials

More platform walkthroughs at https://iot.tertiaryinfotech.com/tutorials

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CONTINUE YOUR LEARNING

Recommended Courses

- WSQ - Internet of Things (IoT) Fundamental for Beginners
- WSQ - Mastering Raspberry Pi: Hands-On Practical Applications for Beginners
- WSQ - Practical Electronics Design with Arduino Microcontroller
- WSQ - Hands-On Guide to IoT Development with Microcontrollers
- Browse the full catalogue at www.tertiarycourses.com.sg

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SUMMARY

Summary & Q&A

- IoT connects physical assets so the business can measure what used to be invisible.
- The opportunity is a business model change, not just a technology deployment.
- Adoption succeeds when value, process integration, capability and security are addressed together.
- Any questions before the assessment?

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ASSESSMENT · 90 MINUTES TOTAL

Final Assessment

WA (Q&A) — 75 min

- Short-answer questions
- Assesses abilities A1–A5
- 15 minutes per ability
- Open book

OQ — 15 min

- Oral questioning
- Assesses knowledge K1–K4
- 1:1 with the assessor
- Open book

Before you begin

- Take the Assessment digital attendance
- Complete the TRAQOM survey
- Slides + Learner Guide allowed
- Submit as directed by your assessor

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit the WA (Q&A) — 75 min, open book
4. Oral Questioning — 15 min, 1:1 with the assessor
5. Sign the Assessment Summary Record

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Digital Attendance & TRAQOM Survey (Mandatory)

1

Assessment attendance

Take the Assessment digital attendance before you begin — scan the SSG QR code.

2

Cert & TRAQOM survey

Complete the mandatory survey at https://lms-tms.tertiaryinfotech.com/

3

Your certificate

Your certificate is issued through the LMS once you are assessed Competent.

4

75% minimum

At least 75% attendance is required to be eligible for assessment and funding.

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WE'RE HERE TO HELP — DURING AND AFTER THE CLASS

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SEE YOU AT THE NEXT COURSE

Thank You!

You can now explore, evaluate and implement business innovation opportunities with IoT.

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